

(2015)

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STAT 641 (Ghosh)

Midterm Exam 1

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Write legibly showing all the steps VERY clearly. Max. score=100 (Total points = 105).

- (1) Define the following: (a) Semi-algebra, (b) algebra, (c) σ -algebra, (d) measurable space, (e) measure, (f) finite measure and probability, (g) σ -finite measure, (h) measure space and probability space, (i) measurable transformation, (j) random variable.

(4 × 10 = 40 points)

- (2) Let $\mathcal{F}$ be a σ -algebra of subsets of $\Omega \neq \emptyset$. Take $\Omega_0 \subset \Omega$ such that $\Omega_0 \notin \mathcal{F}$. Define $\mathcal{F}_0 = \{F \cap \Omega_0 : F \in \mathcal{F}\}$. Show that $\mathcal{F}_0$ is a σ -algebra of subsets of Ω_0 .

(15 points)

- (3) (a) Give an example of a functions f that is not $(\mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}))$ -measurable.
 (b) Give examples of two (distinct) functions f and g that are not $(\mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}))$ -measurable, but $f + g$ is $(\mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}))$ -measurable.
 (c) Let $\Omega_1 = \mathbb{R}$ and $\mathcal{F}_1 = \{\emptyset, (-\infty, 0], (0, \infty), \mathbb{R}\}$ and let $(\Omega_2, \mathcal{F}_2) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Let $T : \Omega_1 \rightarrow \Omega_2$ be defined as

$$T(x) = \begin{cases} 5, & \text{if } x \leq 0 \\ 0, & \text{if } x > 0 \end{cases}$$

Is T $(\mathcal{F}_1, \mathcal{F}_2)$ -measurable? Justify your answer.

(2 × 10 = 20 points)

- (4) Let $\Omega \neq \emptyset$ and $\mu^* : \mathcal{P}(\Omega) \rightarrow [0, \infty]$ be a function ($\mathcal{P}(\Omega)$ is the power set of Ω).
 (a) Define what properties μ^* need to satisfy to be an outer-measure.
 (b) If μ^* is an outer-measure, define what it means for a set $A \subset \Omega$ to be μ^* -measurable.
 (c) Show that if an outer measure μ^* is also finitely-additive, then it is actually a measure on all subsets of Ω (i.e. on $(\Omega, \mathcal{P}(\Omega))$). State any results used in your argument.
 [Hint: start by finding all μ^* -measurable sets]

(3 × 10 = 30 points)

$$\mathcal{M}_{\mu^*} = \{A \subset \Omega : \mu^*(X) = \mu^*(X \cap A) + \mu^*(X \setminus A) \text{ for all } X \subset \Omega\}$$

Let $A \subset \Omega, X \subset \Omega$

$$\begin{aligned} &1 \notin A \\ &0 \in A^c \end{aligned} \quad \begin{aligned} &0 \\ &1 \end{aligned} \quad \begin{aligned} &\in A^c \\ &\in A^c \end{aligned}$$

$$\mu^*(X) = 1$$

$$\Omega \setminus A = (\Omega \setminus F) \cap \Omega$$

$$\cup F_n$$

(2015)

1. (a) $\Omega \neq \emptyset$, $\mathcal{C} \subset \mathcal{P}(\Omega)$ is semi-algebra if
 - (i) $A, B \in \mathcal{C} \Rightarrow A \cap B \in \mathcal{C}$ (π -system);
 - (ii) $\forall A \in \mathcal{C} \Rightarrow A^C = \bigcup_{i=1}^k B_i$, $\{B_i\}_{i=1}^k \subset \mathcal{C}$ are disjoint;
- (b) $\Omega \neq \emptyset$, $\mathcal{F} \subset \mathcal{P}(\Omega)$ is an algebra if
 - (i) $A, B \in \mathcal{F} \Rightarrow A \cup B \in \mathcal{F}$;
 - (ii) $A \in \mathcal{F} \Rightarrow A^C \in \mathcal{F}$;
 - (iii) $\Omega \in \mathcal{F}$.
- (c) $\Omega \neq \emptyset$, $\mathcal{F} \subset \mathcal{P}(\Omega)$ is an σ -algebra if
 - (i) $\{A_i\} \subset \mathcal{F} \Rightarrow \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$;
 - (ii) $A \in \mathcal{F} \Rightarrow A^C \in \mathcal{F}$;
 - (iii) $\Omega \in \mathcal{F}$.
- (d) $(\Omega, \mathcal{F})$ is called a measurable space if $\Omega \neq \emptyset$ and $\mathcal{F}$ is a σ -algebra.
- (e) A set function μ defined on an algebra/semi-algebra/ σ -algebra $\mathcal{F}$ ($\Omega \neq \emptyset$) is called a measure if
 - (i) $\mu : \mathcal{F} \rightarrow [0, \infty]$;
 - (ii) $\mu(\emptyset) = 0$;
 - (iii) $\forall$ disjoint collection of sets $\{A_i\}_i \subset \mathcal{F}$, $\mu(\bigcup_i A_i) = \sum_{i=1}^{\infty} \mu(A_i)$.
- (f) A measure μ is called finite if $\mu(\Omega) < \infty$. A finite measure μ with $\mu(\Omega) = 1$ is called a probability measure.
- (g) A measure μ is σ -finite if $\exists \{A_i, i = 1, 2, \dots\} \subset \mathcal{F}$, s.t. $\bigcup_i A_i = \Omega$ and $\mu(A_i) < \infty, \forall i$.
- (h) If μ is a measure on a measurable space $(\Omega, \mathcal{F})$, then $(\Omega, \mathcal{F}, \mu)$ is called a measure space. In addition, if μ is a probability measure, then $(\Omega, \mathcal{F}, \mu)$ is a probability space.
- (i) If $(\Omega_1, \mathcal{F}_1)$ and $(\Omega_2, \mathcal{F}_2)$ are two measurable spaces and if $X : \Omega_1 \rightarrow \Omega_2$ satisfies $X^{-1}(B) \in \mathcal{F}_1, \forall B \in \mathcal{F}_2$, then X is called a measurable transformation.
- (j) If $(\Omega_1, \mathcal{F}_1)$ has a probability measure defined on it, and if $(\Omega_2, \mathcal{F}_2) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, then the measurable transformation X is called a random variable.

2. Taking $F = \Omega \in \mathcal{F}$, we get $\Omega_0 = \Omega \cap \Omega_0 \in \mathcal{F}_0$.

If $A = F \cap \Omega_0 \in \mathcal{F}_0$ for some $F \in \mathcal{F}$, then $\Omega_0 \setminus A = F^c \cap \Omega_0 \in \mathcal{F}_0$, since $F^c \in \mathcal{F}$ (note that $\Omega_0 \setminus A$ is the complement of A in Ω_0).

If $\{A_i\} \subset \mathcal{F}_0$, then each $A_i = F_i \cap \Omega_0$, for some $\{F_i\} \subset \mathcal{F}$. In this case, $\bigcup_{i \geq 1} F_i \in \mathcal{F}$ and hence $\bigcup_{i \geq 1} A_i = (\bigcup_{i \geq 1} F_i) \cap \Omega_0 \in \mathcal{F}_0$. This proves that $\mathcal{F}_0$ is a σ -algebra of subsets of Ω_0 .

3. (a) $f = 1_A$, where A is a non-Borel measurable set.
 (b) In the same example, take $g = -f$. Then $h = f + g = 0$ is Borel measurable (constant functions are always measurable).
 (c) Note that for $B \in \mathcal{B}(\mathbb{R})$,

$$T^{-1}(B) = \begin{cases} \emptyset; & \text{if } 5 \notin B \text{ and } 0 \notin B \\ (-\infty, 0]; & \text{if } 5 \in B \text{ and } 0 \notin B \\ (0, \infty); & \text{if } 5 \notin B \text{ and } 0 \in B \\ \mathbb{R}; & \text{if } 5, 0 \in B \end{cases}$$

Since $(-\infty, 0], (0, \infty) \notin \mathcal{F}_1$, then T is not $\langle \mathcal{F}_1, \mathcal{F}_2 \rangle$ measurable.

4. (a) $\mu^*(\emptyset) = 0$, $\mu^*(A) \leq \mu^*(B)$ if $A \subset B$ (monotone) and for any countable collection of subsets of Ω , $\{A_i\}$, the following holds $\mu^*(\bigcup_i A_i) \leq \sum_i \mu^*(A_i)$ (ctble. subadditivity).
 (b) A set $A \in \mathcal{P}(\Omega)$ is called μ^* -measurable if: $\mu^*(A) = \mu^*(A \cap E) + \mu^*(A \cap E^c)$ for any $E \in \mathcal{P}(\Omega)$.
 (c) Note that if μ^* is finitely-additive, then $\mu^*(B_1 \cup B_2) = \mu^*(B_1) + \mu^*(B_2)$ for any disjoint B_1, B_2 . Now take any $A \in \mathcal{P}(\Omega)$ and any $E \in \mathcal{P}(\Omega)$. Note that $B_1 = A \cap E$ and $B_2 = A \cap E^c$ are disjoint and hence A is μ^* -measurable. That is, any $A \in \mathcal{P}(\Omega)$ is μ^* -measurable, or in other words, $\mathcal{M}_{\mu^*} =$ set of all μ^* -measurable sets is all of the power set $\mathcal{P}(\Omega)$. But by the theorem in the book (Theorem 1.3.2), μ^* is a measure on $\mathcal{M}_{\mu^*}$ and $\mathcal{M}_{\mu^*} = \mathcal{P}(\Omega)$ in this case. So, μ^* is a measure on $\mathcal{P}(\Omega)$.

1. (a) Let Ω be a nonempty set. Let $\mathcal{C}$ be a class of subsets of Ω . The $\mathcal{C}$ is a semi-algebra if

$$(i) A, B \in \mathcal{C} \Rightarrow A \cap B \in \mathcal{C}$$

$$(ii) A \in \mathcal{C} \Rightarrow \exists B_1, \dots, B_K \text{ for } 1 \leq K < \infty \text{ with } B_1, \dots, B_K \in \mathcal{C} \text{ and } B_i \cap B_j = \emptyset \forall i \neq j \text{ s.t.}$$

$$A^c = \bigcup_{i=1}^K B_i.$$

(b) Let Ω be a nonempty set, and let $\mathcal{F} \subset \mathcal{P}(\Omega)$. Then $\mathcal{F}$ is an algebra if

$$(i) \Omega \in \mathcal{F}$$

$$(ii) A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}$$

$$(iii) A, B \in \mathcal{F} \Rightarrow A \cup B \in \mathcal{F}$$

(c) Let Ω be a nonempty set, and let $\mathcal{F} \subset \mathcal{P}(\Omega)$. Then $\mathcal{F}$ is a σ -algebra if $\mathcal{F}$ is an algebra and

$$(iv) \text{ if } \{A_n\}_{n=1}^\infty \subset \mathcal{F},$$

$$\bigcup_{n=1}^\infty A_n \in \mathcal{F} \quad (\text{closed under countable unions})$$

(d) $(\Omega, \mathcal{F})$ is a measurable space if Ω is a nonempty set and $\mathcal{F}$ is a σ -algebra

(e) Let Ω be a non-empty set, and let $\mathcal{F}$ be an algebra on Ω . Let μ be a set function on $\mathcal{F}$. Then μ is a measure if

$$(i) \mu(A) \in [0, \infty] \quad \forall A \in \mathcal{F}.$$

$$(ii) \mu(\emptyset) = 0$$

$$(iii) \text{ if } \{A_n\}_{n=1}^\infty \subset \mathcal{F} \text{ with } A_i \cap A_j = \emptyset \forall i \neq j \text{ and } \bigcup_{n=1}^\infty A_n \in \mathcal{F}, \text{ then } \mu\left(\bigcup_{n=1}^\infty A_n\right) = \sum_{n=1}^\infty \mu(A_n).$$

(f) Let μ be a measure. Then μ is finite if $\mu(\Omega) < \infty$;
and μ is a probability if $\mu(\Omega) = 1$.

(g) Let $\mathcal{F}$ be a σ -algebra and μ be a measure on $\mathcal{F}$.

Then μ is σ -finite if

(i) $\exists \{A_n\}_{n=1}^{\infty} \subset \mathcal{F}$ with $\bigcup_{n=1}^{\infty} A_n = \Omega$ and

(ii) $\mu(A_n) < \infty \quad \forall n \geq 1$.

(h) A measure space is a triplet $(\Omega, \mathcal{F}, \mu)$ where Ω is a non-empty set, $\mathcal{F}$ is a σ -algebra, and μ is a measure on $\mathcal{F}$. Additionally, if μ is a probability measure, then $(\Omega, \mathcal{F}, \mu)$ is probability space.

(i)

(j)

(2) $\mathcal{F}$ a σ -algebra with $\mathcal{F} \subset \mathcal{P}(\Omega)$ for $\Omega \neq \emptyset$.

$$\Omega_0 \subset \Omega \text{ s.t. } \Omega_0 \notin \mathcal{F}$$

$$\mathcal{F}_0 = \{F \cap \Omega_0 : F \in \mathcal{F}\}$$

Want to show that $\mathcal{F}_0$ is a σ -algebra of subsets of Ω_0 .

We 1st show $\Omega \in \mathcal{F}_0$.

Consider that $\Omega \in \mathcal{F}$ since $\mathcal{F}$ a σ -algebra.

Thus

$$\Omega \cap \Omega_0 = \Omega_0 \in \mathcal{F}_0. \checkmark$$

We now want to show $A \in \mathcal{F}_0 \Rightarrow A^c \in \mathcal{F}_0$.

Let $A \in \mathcal{F}_0$. Then $A = F \cap \Omega_0$ s.t. $F \in \mathcal{F}$.

WTS $A^c = (F \cap \Omega_0)^c = F^c \cup \Omega_0^c \in \mathcal{F}_0$. Additionally, consider that

$$\begin{aligned} A^c &= \Omega_0 \setminus A = \Omega_0 \cap A^c = \Omega_0 \cap (F^c \cup \Omega_0^c) \\ &= \Omega_0 \cap (F^c \cup \emptyset) \\ &= \Omega_0 \cap F^c \end{aligned}$$

We know that $F^c \in \mathcal{F}$, since $F \in \mathcal{F}$ and $\mathcal{F}$ a σ -algebra.

Thus

$$\Omega_0 \cap F^c \in \mathcal{F}_0. \checkmark$$

$$\Rightarrow A^c \in \mathcal{F}_0.$$

Finally, we want to show that for any sequence $\{A_n\}_{n \geq 1} \subset \mathcal{F}_0$, that $\bigcup_{n \geq 1} A_n \in \mathcal{F}_0$.

$$\bigcup_{n \geq 1} A_n = \bigcup_{n \geq 1} (F_n \cap \Omega_0)$$

$$= \left(\bigcup_{n \geq 1} F_n \right) \cap \Omega_0$$

Since $\mathcal{F}$ a σ -algebra.

$$\begin{matrix} F_n \in \mathcal{F} \\ \forall n \geq 1 \end{matrix} \quad \bigcup_{n \geq 1} F_n \in \mathcal{F}.$$

Hence,

$$\bigcup_{n \geq 1} A_n = \left(\bigcup_{n \geq 1} F_n \right) \cap \Omega_0 \in \mathcal{F}_0.$$

(4) We have $\Omega \neq \emptyset$ and $\mu^*: \mathcal{P}(\Omega) \rightarrow [0, \infty]$ a function, $\therefore$

(a) In order for μ^* to be an outer measure, μ^* needs to have the following properties:

(i) $\mu^*(\emptyset) = 0$

(ii) (monotonicity) If $A \subset B$ with $A, B \in \mathcal{P}(\Omega)$, then $\mu^*(A) \leq \mu^*(B)$.

(iii) (countable subadditivity) Let $\{A_n\}_{n=1}^\infty$ be any sequence in $\mathcal{P}(\Omega)$. Then

$$\mu^*\left(\bigcup_{n=1}^\infty A_n\right) \leq \sum_{n=1}^\infty \mu^*(A_n).$$

(b) Suppose μ^* is an outer-measure. Let A be a set s.t. $A \subset \Omega$. In order for A to be μ^* -measurable, it must be for any set $E \subset \Omega$,

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c).$$

(c) Let the outer measure μ^* be finitely-additive. We want to show that it is actually a measure on all subsets of Ω .

Let A be any set s.t. $A \in \mathcal{P}(\Omega)$. Also, let E be any set s.t. $E \in \mathcal{P}(\Omega)$. Then, consider that

$$E = (A \cap E) \cup (A^c \cap E),$$

and $(A \cap E) \cap (A^c \cap E) = \emptyset$ are disjoint sets. Since we are assuming that μ^* is finitely-additive,

$$\mu^*(E) = \mu^*((A \cap E) \cup (A^c \cap E)) = \mu^*(A \cap E) + \mu^*(A^c \cap E).$$

Hence, A is μ^* -measurable. Since A was arbitrary, all sets in $\mathcal{P}(\Omega)$ are μ^* -measurable. By Carathéodory's Thm (not ext), for any outer measure μ^* and $\mathcal{M} = \{A \mid A \text{ is } \mu^*\text{-measurable}\}$, μ^* restricted to $\mathcal{M} = \mathcal{P}(\Omega)$ is a measure!

① (h) The Ω be a nonempty set and $\mathcal{F}$ a σ -algebra on Ω . Furthermore, let μ be a measure on $(\Omega, \mathcal{F})$. Then

$(\Omega, \mathcal{F}, \mu)$ is a measure space. If $\mu = P$ is a probability measure, then $(\Omega, \mathcal{F}, P)$ is a probability space.

(i) Let $(\Omega_1, \mathcal{F}_1)$ and $(\Omega_2, \mathcal{F}_2)$ be two measure spaces. Let T be a function from Ω_1 to Ω_2 . Then if

$$T^{-1}(A) \in \mathcal{F}_1 \quad \forall A \in \mathcal{F}_2,$$

then T is a $(\mathcal{F}_1, \mathcal{F}_2)$ -measurable transformation.

(j) Let X be a function from $(\Omega_1, \mathcal{F}_1, P)$ to $\mathbb{R}$.

Then X is a random variable if

$$X^{-1}(A) \in \mathcal{F}_1 \quad \forall A \in \mathcal{B}(\mathbb{R}).$$

③ (a) Let B be a non-Borel measurable set, then

$f = I_B$ is not $(\mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}))$ -measurable.

(b) Let $f = I_B$ and $g = -I_B$. Then $h = f + g = f - f = 0$, and constant functions are always measurable.

(c) Consider that for $A \in \mathcal{F}_2$

$$T^{-1}(A) = \begin{cases} \emptyset & \text{if } 0 \notin A \text{ and } 5 \notin A \\ (-\infty, 0] & \text{if } 0 \notin A \text{ and } 5 \in A \\ (0, \infty) & \text{if } 0 \in A \text{ and } 5 \notin A \\ \mathbb{R} & \text{if } 0 \in A \text{ and } 5 \in A \end{cases}$$

All of these values are in $\mathcal{F}_1$, so T is $(\mathcal{F}_1, \mathcal{F}_2)$ -measurable.